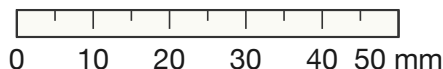


Perturbation Placement Guide — 150 mm Rod (1:1 Scale)

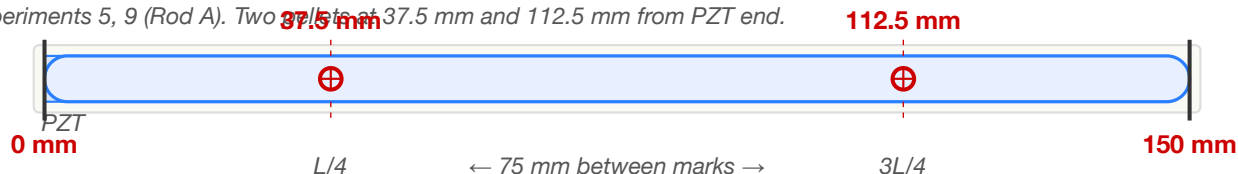
Print at 100% (no scaling). Lay your glass rod along each blue channel to mark putty positions.

Verify: the calibration ruler below should measure exactly 50 mm with a physical ruler.



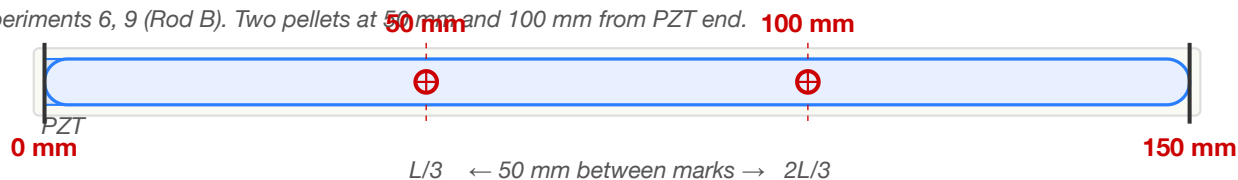
PATTERN A — Quarter-points: $L/4$ and $3L/4$

Experiments 5, 9 (Rod A). Two pellets at 37.5 mm and 112.5 mm from PZT end.



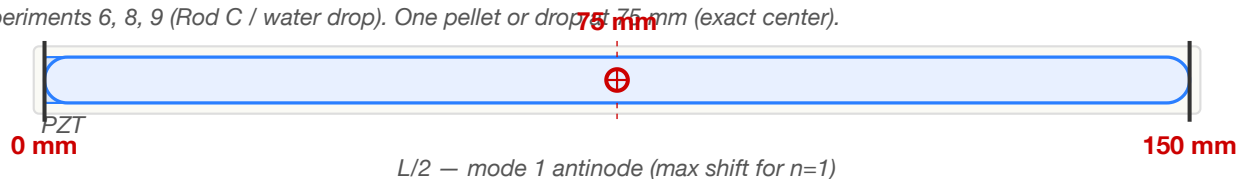
PATTERN B — Third-points: $L/3$ and $2L/3$

Experiments 6, 9 (Rod B). Two pellets at 50 mm and 100 mm from PZT end.



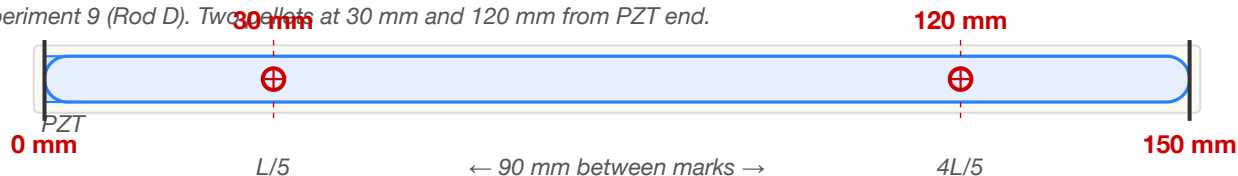
PATTERN C — Midpoint: $L/2$

Experiments 6, 8, 9 (Rod C / water drop). One pellet or drop at 75 mm (exact center).



PATTERN D — Fifth-points: $L/5$ and $4L/5$

Experiment 9 (Rod D). Two pellets at 30 mm and 120 mm from PZT end.



REFERENCE RULER — All perturbation positions (150 mm rod)

All positions from Experiments 5–11 on a single ruler. Tick labels show position and fraction.

